

Proposal for developing customized microbial fertilizers using biomass

1. Executive Summary

We are developing **single-use, crop-specific fertilizers** through **biomass**, converting an industrial byproduct into a sustainable agricultural input. Our solutions are designed for a wide range of crops — from wheat, paddy, and sugarcane to pulses, spices, fruits, and vegetables — offering farmers an affordable and eco-friendly alternative to chemical fertilizers.

At the heart of this technology lies a fusion of **biotechnology, nanotechnology, and chemical engineering**. Using an advanced combination of **microorganisms, nanoparticles, lignin-based superabsorbent polymers, and bioreactor-based processing**, we enhance the nutrient composition of FOM with balanced **Nitrogen, Phosphorus, and Potassium (NPK)**, high organic carbon, and improved microbial activity.

These bio-nano fertilizers deliver measurable impact — up to **21% reduction in farming costs, 15–35% increase in yield, and 33% reduction in irrigation requirements**. Beyond yield, they strengthen soil microbial diversity, improve water retention, and promote healthy root growth, leading to long-term soil rejuvenation.

Our technology has been **tested and certified by the Indian Council of Agricultural Research (ICAR)** for its efficiency, economy, and contribution to natural farming. We are collaborating with organizations such as **BCPL, HPCL, KSK Energy, Carbon Circle**, and others to repurpose agro-industrial byproducts into nutrient-rich fertilizers. Supported by the **Government of Madhya Pradesh** and the **Principal Scientific Adviser to the Government of India**, the initiative aims to establish **100+ decentralized mini-manufacturing units**, creating rural employment, improving waste management, and restoring soil health.

In partnership with **Tata Trusts, the Bill & Melinda Gates Foundation, TRIF, Pradan**, and multiple livelihood organizations, our mission extends beyond innovation — to make agriculture more resilient, circular, and profitable for every farmer.

2. Objectives:

- Utilization of low nutrient value biomass which is having low nutrition value as a carrier for microbial consortia to develop crop specific fertilizers using advance technology.
- To achieve a reduction of 20% to 25% in the overall farming expenditure, enhance crop yields by 15% to 35%, increase microorganism populations, C/N ratio and humas content in the soil.
- Minimization of ground water wastage by at least 33% in very first crop cycle by using super absorbent polymer developed using agricultural biomass known as parali which can absorb 268 times water as compare to its weight
- To promote organic and sustainable farming practices for cash crops by offering a 100% organic product range. This initiative aims to reduce the presence of chemical residues in both soil and crop produce.
- To create employment opportunities for both skilled (5) and unskilled (25+) laborers while actively promoting women's empowerment (35%) within society through each production unit.

3. Background and Motivation:

In the context of Indian agriculture, the use of chemical fertilizers has given rise to numerous challenges. The application of these fertilizers results in the release of harmful gases, contributing to at least 11,000 deaths annually. Moreover, the heavy application of chemical products leads to soil pH alterations, causing a significant decline in soil microorganism populations and water retention capacity. This, in turn, results in the utilization of approximately 85% of groundwater for irrigation in Indian agriculture.

Microorganisms play a crucial role in breaking down complex nutrient molecules into a format readily absorbed by plants. A reduction in microorganism populations increases the demand for complex chemical fertilizers. Furthermore, more than 80%+ of these chemical products remain unused in the soil, finding their way into the human body and causing various health-related issues, including the alarming rise in cancer cases.

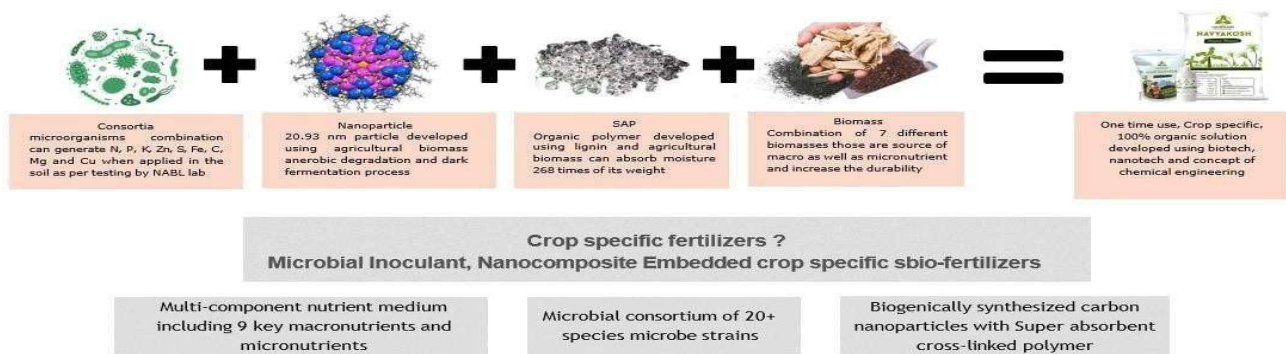
The escalating demand for fertilizers, insecticides, and irrigation exacerbates the economic challenges faced by farmers, leading to substantial losses. According to the National Crime Records Bureau (NCRB) report, India witnesses over 3,500 farmer suicides annually because of these direct circumstances.

4. Conventional solutions

There are three primary categories of agricultural inputs available in the market:

- Chemical fertilizers such as Urea, DAP, NPK, and Zinc Sulphur. These fertilizers supply essential nutrients to the soil and are applied periodically during farming. However, their use is associated with significant environmental issues, including alterations in soil pH, soil depletion, and a decline in organic matter and humus content. These factors collectively lead to an increase overall farming expenditure due to heightened demand for fertilizers and irrigation.
- Organic manure, including vermicompost and other organic materials, enrich the soil with humus and nutrients. However, achieving average crop yields often requires the use of chemical products in conjunction with these organic inputs due to the need for substantial quantities of organic materials.
- Biofertilizers such as mycorrhiza, Azotobacter, and Rhizobia are also available. These biofertilizers tend to be relatively expensive, and their use typically necessitates the simultaneous application of both organic manure and chemical fertilizers to attain average crop yields. Consequently, this practice disrupts the economic sustainability of agriculture.

The application of chemical fertilizers can diminish soil fertility by affecting its pH, resulting in a reduced microorganism count. This, in turn, leads to heightened demand for fertilizers, ongoing irrigation requirements, and suboptimal crop yields.



5. Novelty

We typically categorize our technology into three distinct domains: biotechnology, nanotechnology, and engineering concepts. This classification allows us to develop a range of agricultural products that are one-time use, crop-specific, and entirely organic, tailored to the specific needs of plants and prevailing climatic conditions.

i. Biotechnology Part

In this process, we isolate and combine 18 different types of microorganisms responsible for producing nine distinct factors essential for plant growth. These factors encompass Nitrogen, Phosphorus, Potash, Zinc, Carbon, Sulphur, Iron, and more, ensuring a comprehensive nutrient supply throughout the crop cycle from the initial application.

To achieve this, we isolate microorganisms from diverse sources, including forests, agricultural lands, ponds, and industrial settings, with the goal of creating a consortium capable of delivering the necessary nutrition to plants. These microbes not only fulfill the plant's nutrient requirements but also contribute to soil health enhancement by increasing organic matter content.

Furthermore, we formulate combinations of eight different types of biomass, in which we use old cow dung as a major raw material along with this we also use some quantity of jaggery, and alcohol industries, waste from dairy and rice mills, agricultural residues, and agricultural products. These biomass sources serve as nutrient and micronutrient reservoirs and act as suitable carriers for microorganisms.

Our research extends to extensive field trials conducted in various regions of Uttar Pradesh, Bihar, and Jharkhand. These trials evaluate the impact of our formulations on different crops. Additionally, we have monitored the year-to-year increase in organic matter within the soil, and our findings indicate a substantial 42% increase, underscoring the positive effects of our approach.

ii. Nanotechnology

In this segment, we have organically synthesized nanoparticles designed to provide an expanded surface area for microorganisms. These nanoparticles offer several additional advantages, including enhancing the thermophilic properties of microorganisms up to 78 degrees Celsius. They also facilitate the decomposition of available biomass, leading to increased organic matter content and ultimately contributing to the augmentation of microorganism populations.

To achieve this, we conducted comprehensive experiments focusing on five critical factors: pH levels, temperature conditions, the number of fermentation days, the type of fermentation, and, finally, calcination or roasting. Our systematic experimentation yielded optimal results

iii. Chemical engineering

To expedite the manufacturing process of our crop-specific biofertilizers and insecticides, we have undertaken the development of low cost customized fermenter tanks. These bioreactors are designed to allow precise control over various climatic conditions, including temperature, humidity, pH levels, and agitation, facilitating anaerobic solid-state fermentation.

Additionally, we are actively working on the creation of an organic superabsorbent polymer or hydrogel. This polymer exhibits the remarkable ability to absorb water molecules up to 268 times its weight, releasing water slowly into the soil over a period of one month. Furthermore, due to its organic nature, these polymers

naturally degrade. Their incorporation not only addresses irrigation concerns but also alleviates plant stress factors such as relative electrical conductivity, proline levels, MDA (Malondialdehyde) content, hydrogen peroxide, and POD (Peroxidase) activity. By combining these innovations and applying them to seeds, we aim to resolve multiple farming challenges while providing the benefits of each component in a unified solution.

Our research has demonstrated that this approach can lead to yield increases ranging from 15% to 40% across different crops. It also substantially reduces irrigation demands by at least 33%, decreases farmers' investments by 45%, and minimizes the effort required, as it is a one-time application that obviates the need for additional fertilizers and micronutrients throughout the crop cycle. Furthermore, our solution reduces the crop duration by 5-8%, as confirmed through comparative analyses with chemical fertilizers.

6. Technology and product details

Navyakosh represents a new generation of smart bio-nano-polymer-based fertilizers developed by LCB Fertilizers Pvt. Ltd. through the convergence of biotechnology, nanotechnology, and chemical engineering. The initiative focuses on creating six distinct varieties of Navyakosh, each scientifically tailored through combinations of microorganisms that respond to three fundamental parameters — soil type, crop category, and climatic conditions. This innovation aims to transform fertilizer design from a one-size-fits-all approach to a precision-based, adaptive system that ensures maximum nutrient efficiency, soil health improvement, and resilience under diverse environmental challenges.

Scientific Foundation: Soil, Crop, and Climate Optimization

The development of Navyakosh begins with detailed hyperspectral analysis of soil to identify its physicochemical and biological characteristics. Using advanced spectral mapping, parameters such as nitrogen availability, organic matter, moisture retention, and vegetation indices are assessed. The findings from these analyses reveal that fields treated with Navyakosh show significantly improved vegetation health, chlorophyll content, and moisture retention compared to conventional fields.

High NDVI and NDRE values indicate denser canopy development and greater photosynthetic activity, while elevated NDMI readings confirm enhanced water-holding capacity. These studies form the diagnostic backbone of Navyakosh formulations, allowing each variant to be custom-designed based on the real needs of the soil and the crops cultivated in specific regions. The result is a highly adaptive and data-driven formulation that supports efficient nutrient cycling and long-term soil fertility restoration.

Parameter / Index	Full Form / Description	What It Measures	Interpretation Range	Practical Application in Agriculture
NDVI	<i>Normalized Difference Vegetation Index</i>	Measures vegetation greenness and vigor. Indicates plant health and canopy density.	0.60–1.00: Dense vegetation 0.40–0.59: Moderate 0.20–0.39: Low 0.10–0.19: Very sparse 0.00–0.09: Bare soil	Used to monitor crop growth stages, detect stress, and estimate overall crop health and yield potential.

MSAVI	<i>Modified Soil Adjusted Vegetation Index</i>	Adjusts NDVI by reducing soil background effect, especially in sparse vegetation areas.	Similar to NDVI, but more stable under low canopy cover.	Helpful in early crop stages or low-density crops where soil background affects readings.
RECI	<i>Red Edge Chlorophyll Index</i>	Estimates chlorophyll concentration and plant photosynthetic activity.	0.45–1.00: Very good 0.35–0.44: Good 0.25–0.34: Moderate 0.15–0.24: Low <0.15: Bare/dead vegetation	Detects nutrient deficiencies (e.g., nitrogen), identifies stress zones, and supports fertilizer optimization.
NDRE	<i>Normalized Difference Red Edge Index</i>	Similar to RECI but better suited for dense canopies and later crop stages.	Same as RECI classification ranges.	Allows early detection of plant stress before it is visible, used for canopy analysis and nitrogen management.
NDMI	<i>Normalized Difference Moisture Index</i>	Measures leaf and soil moisture content. Detects early water stress.	0.40–1.00: Very high moisture 0.30–0.39: Good 0.20–0.29: Moderate 0.10–0.19: Low <0.10: Very low	Used for irrigation scheduling, drought monitoring, and soil-water retention analysis.
EVI	<i>Enhanced Vegetation Index</i>	Measures vegetation density with higher sensitivity in high biomass areas.	0.60–1.00: Dense 0.30–0.59: Moderate 0.00–0.29: Sparse	Complements NDVI in dense vegetation zones; useful for yield estimation and long-term vegetation studies.
LAI	<i>Leaf Area Index</i>	Quantifies leaf coverage per unit ground area. Indicates canopy size and	Higher LAI → denser foliage and better productivity.	Used in yield estimation models like WOFOST; helps correlate biomass with productivity.

WOFOST Model	<i>World Food Studies Crop Growth Model</i>	Simulates crop physiological processes such as photosynthesis and water use.	Uses multiple vegetation indices and weather parameters.	Provides early yield forecasts (up to 2 months), supporting planning and input
Slope & Elevation	Derived from Digital	Analyze topographic	—	Used to study moisture
Maps	Elevation Models (DEM)	effects on water runoff and soil moisture.		retention, erosion risk, and crop suitability based on terrain gradient.
Nitrogen, Phosphorus, Potassium Maps (NPK Maps)	Derived using NDVI and NDRE indices	Show spatial distribution of essential nutrients in soil.	—	Used for variable rate fertilization and soil nutrient management planning.
Productivity Maps	Composite of NDVI, NDRE, MSAVI, NDMI	Represent multi-layered yield potential zones.	—	Identify high- and low-productivity zones for targeted agronomic interventions.
Sowing Maps	Based on NDVI temporal variation	Identify crop establishment and germination zones.	—	Used to monitor sowing success, crop stand uniformity, and identify re-sowing needs.

Microbial Engineering and Formulation Design

The microbial composition of Navyakosh is a carefully engineered consortium of beneficial microorganisms that work synergistically to enhance soil fertility and crop yield. The formulation includes phosphate-solubilizing bacteria, Azospirillum, Azotobacter, potassium-mobilizing bacteria, mycorrhiza, Pseudomonas fluorescens, Azolla, and other plant growth-promoting rhizobacteria. Each strain plays a unique role — fixing atmospheric nitrogen, solubilizing locked-up phosphorus and potassium, mobilizing micronutrients, and improving root health.

The selection and ratio of microorganisms vary depending on the soil and crop requirements. For instance, nitrogen-fixing microbes are emphasized in cereal-based formulations, while phosphate solubilizers and mycorrhiza dominate in legume and horticulture formulations. These microbes rejuvenate the soil's biological ecosystem, increase organic matter, and enable crops to utilize macro and micronutrients efficiently throughout the growing cycle.

Nano-Biotechnology Integration

To enhance microbial performance, the formulation incorporates green-synthesized nanoparticles that act as carriers and protectants. These nanoparticles provide a greater surface area for microbial adhesion and



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nutrient exchange while ensuring survival in extreme environmental conditions, even up to 78°C. They create a microenvironment that shields the microbial culture from desiccation, UV exposure, and temperature fluctuations.

This integration of nanotechnology not only enhances microbial efficiency but also enables targeted nutrient delivery to plant roots. The nanoscale interactions improve nutrient solubility and uptake while allowing longer microbial persistence in the rhizosphere. This innovation bridges the gap between traditional

biofertilizers and advanced climate-resilient agricultural inputs.

Polymer Matrix for Moisture Management

Each microbial-nanoparticle composite is encapsulated within a lignin-based superabsorbent polymer developed through LCB's proprietary process. This biopolymer has a remarkable ability to absorb water up to 268 times its weight and retain soil moisture for more than 35 days. It gradually biodegrades within six months, enriching the soil's organic carbon and maintaining ecological balance.

The polymer acts as a natural moisture reservoir, ensuring that both the plant roots and the microbial population have sustained access to water even under drought conditions. This reduces crop stress, minimizes irrigation demand, and supports the continuous release of nutrients. It also improves soil structure, reduces erosion, and contributes to a more porous and aerated soil profile, ultimately promoting root elongation and microbial activity.

Production Process and Fermentation System

Navyakosh is delivered in a concentrated form that can be expanded into 2000 kilograms of ready-to-use fertilizer using LCB's specially designed fermentation chambers located across different regions. The process utilizes locally available raw materials such as low-nutrient-value FOM, rock phosphate, green sand, chicken litter, and other organic additives to maintain cost-effectiveness and sustainability.

Each fermentation chamber is equipped with automated controllers designed by LCB to regulate temperature, humidity, and pH, ensuring optimal microbial growth and nutrient conversion. The fermentation process accelerates the breakdown of organic substrates, transforming them into a nutrient-rich biofertilizer within seven to ten days. This decentralized production model empowers local Farmer Producer Organizations (FPOs) and Self-Help Groups (SHGs) to manufacture Navyakosh locally, reducing transportation costs and supporting rural entrepreneurship.

Nutrient Composition and Agronomic Performance



ICAR-IIPR

Comparative analysis in Chemical, organic, chemical+ organic, control and Navyakosh

9	33%	20-30%	15-35%	34%	15%	5%	15-20%	15%	10-20%
Nutrients generating microorganisms	Reduction in irrigation demand	Reduction in input cost	Increase in agricultural yield	Improvement in soil C/N ratio	Higher chlorophyll content in plants	Reduction in days of maturity of plants	Higher nutrition value of the soil	Higher nutrition value of the grains	Higher monetary returns to farmers

NABL Accredited Lab

Nutrient analysis of Navyakosh containing N, P, K, Zn, S, Fe, C, Mg, Cu, B



DAV-PG College

Comparative analysis of Navyakosh with chemical fertilizers analyzing profitability

As per the comparative analysis done by Indian Council of Agriculture Research (ICAR)

The final Navyakosh product delivers a balanced nutrient profile with approximately 8% nitrogen, 6% phosphorus, and 6% potassium, along with 30% organic carbon. It also contains vital micronutrients including zinc, sulphur, iron, magnesium, and boron. This combination enhances nutrient availability, boosts microbial count, and promotes sustained soil fertility.

Field validation studies conducted by the Indian Council of Agricultural Research (ICAR) confirmed the effectiveness of Navyakosh compared to chemical, organic, and integrated fertilizer systems. Trials showed a 15–35% increase in yield, up to 48% reduction in overall agricultural input costs, and a 33% reduction in irrigation requirements. Navyakosh-treated soils displayed improved organic carbon levels, better moisture retention, and higher microbial density — proving its potential as a sustainable and scalable solution for modern agriculture.

7. ICAR Certification Overview

The bio-nano fertilizer “Navyakosh”, developed by **LCB Fertilizers Pvt. Ltd.**, has been **tested and certified by the Indian Council of Agricultural Research (ICAR)** for its effectiveness, safety, and contribution to sustainable agriculture. The evaluation was carried out by the **Division of Crop Production, ICAR–Indian Institute of Pulses Research (IIPR), Kanpur** during the **Kharif season of 2023**.

This certification affirms that *Navyakosh* meets the agronomic, economic, and environmental benchmarks required for large-scale agricultural use. The trials, conducted under controlled field conditions, compared *Navyakosh* with standard chemical fertilizers, farmyard manure, and integrated nutrient management practices in **mungbean (*Vigna radiata*)** cultivation. The results confirmed that *Navyakosh* offers superior performance across key growth, yield, and soil health parameters.

The product demonstrated **enhanced nutrient use efficiency, improved soil organic matter, and a significant rise in yield and profitability**, validating its position as a next-generation fertilizer technology.

Certification Summary Table: Comparative Performance Analysis

Category	Parameter / Indicator	Navyakosh (375 kg/ha)	Chemical Fertilizer (20:60:40 NPK)	Control	Key Observation
Crop Type	Mungbean (<i>Vigna radiata</i>)	✓	✓	✓	Field evaluation conducted under Kharif conditions on sandy loam soil (pH 8.2).
Plant Growth	Plant Height	58.2 cm	57.7 cm	46.7 cm	Increased vegetative growth by ~24% over control.
	Branches/Plant	2.42	2.37	2.17	Improved canopy structure and branching.

	Leaves/Plant	27.5	27.0	24.0	14–15% increase in leaf density.
	Leaf Length	6.96 cm	6.83 cm	5.89 cm	Higher chlorophyll area index supports better photosynthesis.
Root & Nodule Parameters	Root Fresh Weight	9.73 g	9.64 g	7.26 g	34% higher than control, indicating stronger root development.
	Root Dry Weight	1.26 g	1.23 g	1.11 g	Enhanced nutrient absorption and soil anchorage.
Physiological Traits	Chlorophyll Content	40.7 mg/g	39.1 mg/g	35.4 mg/g	14.9% higher chlorophyll content ensures better photosynthetic efficiency.
	Flowers/Plant	27.1	26.6	22.3	21.5% higher flowering rate — better reproductive performance.
	Days to Maturity	60.8	65.0	62.8	Early maturity by 2–4 days; favorable for short-duration crops.
Yield Attributes	Pods/Plant	20.3	20.0	16.7	21.6% higher pod formation per plant.
	Pod Length	7.53 cm	7.39 cm	6.21 cm	Larger pod size with uniform seed filling.
	100-Seed Weight	4.03 g	3.95 g	3.23 g	24.8% increase in test weight, improving grain quality.
	Grain Yield	7.51 q/ha	7.16 q/ha	5.82 q/ha	29% higher yield over control; 5% higher than chemical fertilizer.
Soil Health	Available Nitrogen	103.5 kg/ha	94.1 kg/ha	97.2 kg/ha	Improved nitrogen fixation and microbial activity.
	Available Phosphorus	14.4 kg/ha	11.0 kg/ha	10.3 kg/ha	30.9% higher P availability — enhanced

					solubilization efficiency.
	Available Potassium	230.3 kg/ha	178.9 kg/ha	165.1 kg/ha	28.7% increase in soil K level after crop harvest.
	Organic Carbon	0.31%	0.31%	0.24%	Sustained organic matter content and humus improvement.
Nutrient Composition in Plant	Nitrogen (%)	2.31	2.15	2.09	Improved nutrient uptake efficiency.
	Phosphorus (%)	0.28	0.26	0.20	Higher P assimilation recorded in treated plants.
	Potassium (%)	1.15	1.13	1.01	Efficient mobilization and translocation.
Seed Nutrient Quality	Nitrogen (%)	3.73	3.66	3.42	Increased protein content and nutritional quality.
	Phosphorus (%)	0.50	0.48	0.43	16.3% improvement over control.
	Potassium (%)	0.73	0.70	0.65	Maintained higher K concentration for seed vigor.
Economic Indicators	Net Return (₹/ha)	38,586	34,278	28,613	12.6% higher net returns over chemical fertilizer.
	B:C Ratio	1.50	1.27	1.35	18.1% higher benefit-cost ratio; proven economic viability.

Key Findings and Certification Highlights

- **ICAR Validation:** The field trials scientifically confirmed *Navyakosh* as a high-performance, crop-safe fertilizer that significantly enhances mungbean productivity.
- **Yield Advantage:** *Navyakosh* delivered up to **29% higher yield** compared to control and **5% higher than chemical fertilizers**, ensuring consistent crop response.
- **Soil Restoration:** Post-harvest soil analysis recorded higher levels of nitrogen, phosphorus, and potassium, along with improved microbial activity and stable organic carbon.
- **Water and Resource Efficiency:** Due to enhanced root health and soil structure, the fertilizer contributed to **better water-use efficiency** and early crop maturity.
- **Economic Viability:** A **Benefit:Cost ratio of 1.50** and **12.6% higher net returns** establish *Navyakosh* as both an agronomically superior and financially sustainable input.

- **Sustainability and Safety:** The ICAR certification positions *Navyakosh* as a scalable alternative to chemical fertilizers, promoting natural farming and long-term soil fertility.

Certification Statement

The product “*Navyakosh*”, developed by **LCB Fertilizers Pvt. Ltd.**, stands **certified by ICAR** for its efficacy in improving crop yield, soil nutrient balance, and economic returns. The certification validates that *Navyakosh* meets the national standards for **organic, safe, and sustainable fertilizer technology**, making it a **reliable alternative to conventional inputs** and a **key enabler of next-generation green agriculture**.

8. Financial analysis of developed product

Commercial analysis of produced fertilizer is done on production capacity of 25 ton/ month in which raw material like cow dung, Jaggery, Microbial consortium, and gram floor are used. In this process a proper ratio is to be developed as mentioned in sheet which is further added in tank (capacity 2.1 ton). In the tank climatic conditions like temperature, pH and humidity is to be maintained using suitable climatic condition controller as per the requirement of microbial consortium.

Raw Material	Rupees/Kg	weight per tank	Cost/tank
Cow dung	1	2000	2000
jaggery	18	5	90
microbial consortium	4000	1	4000
Rock phosphate	6	200	1200
Gram floor	60	5	300

Table 4- Cost/ tank of raw material

The process of fermentation will occur inside the tank and along with suitable climatic condition dark fermentation and anerobic degradation process will occur for 7 to 10 days. After the process of degradation the produced material will be sent for packaging.

Raw material/ kg fertilizer			
Composition of raw material	Fertilizer produced	Cost	Cost/Kg
Tank	2000	7950	3.80

Table 5- Cost/ kg of raw material

Working cost		
operation cost per month	Rs/ month	Per kg
worker charges 3 worker	24000	0.96
Plant maintenance	3000	0.12
Electricity	2500	0.10
Packaging (20 Rs per bag)	10000	0.40
total	39500	1.58

Table 6- working cost

Other Cost		
Targeting 2000 packet (100k kg)/month	Charges/ month	Rs/Kg
Marketing and advertisement	10000	0.40
Additional cost	5000	0.20
Total	15000	0.60

Table 6- Other cost

Production Cost		
Weight	1Kg	50 kg
cost	5.98	299.00

Table 7- Final production cost/bags

Margin	%	Rs for 50kg
MRP		850
Selling Price		450

total bags	500
bags weight (kg)	50
Total production/ month	25000
Total operation cost	149500
Total revenue	225000
Total profit	112375

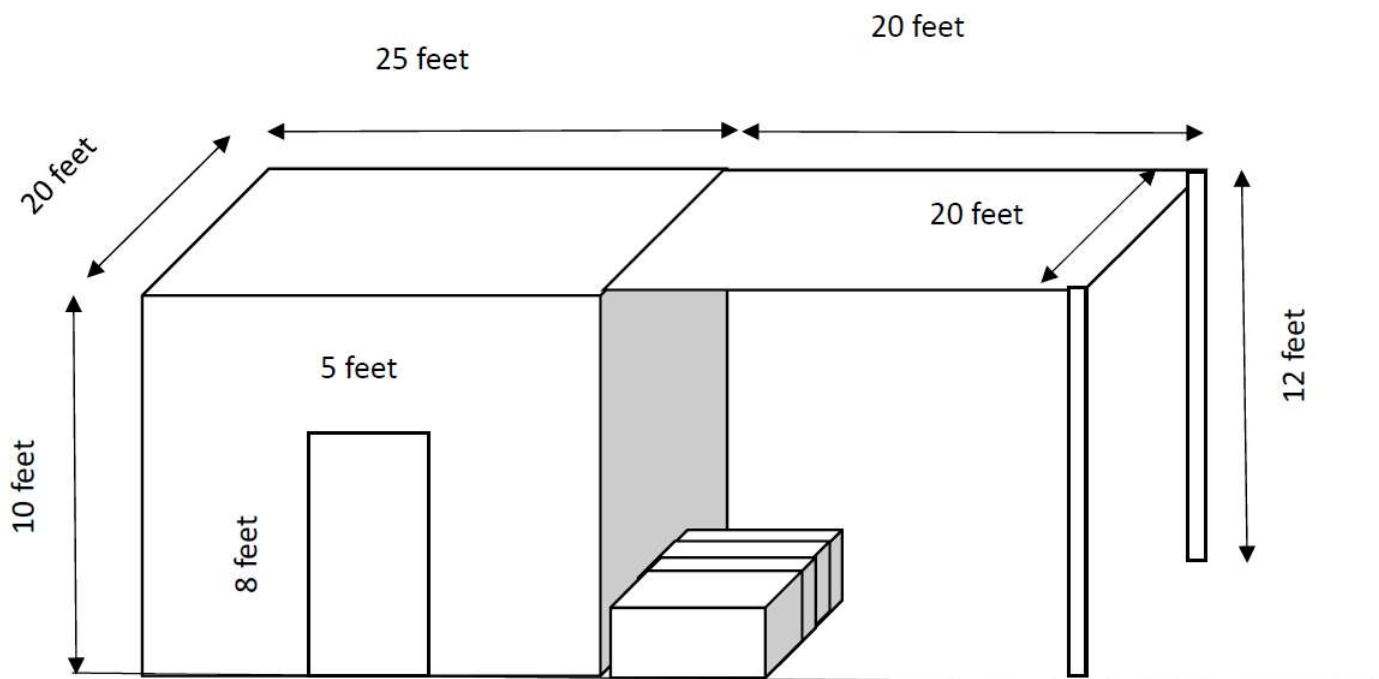
Table 9- Total profit

The commercial analysis of the produced fertilizer considers a production capacity of 25 tons per month, utilizing raw materials such as FOM, jaggery, microbial consortium, cow dung, and gram flour. The process requires developing a specific ratio as detailed in the sheet, which is then added to a tank with a 2.1-ton capacity. The tank's climatic conditions, including temperature, pH, and humidity, are controlled to meet the microbial consortium's requirements. Fermentation occurs within the tank, involving dark fermentation and anaerobic degradation for 7 to 10 days. After degradation, the material is packaged. Financially, cow dung, costing Rs 1/kg, can be commercialized at Rs 12-16/kg.

9. Cost for setting the infrastructure

Infrastructure	Capacity	Number	Cost
Incubator	15 culture	1	85000
Sensors		4	25000
pH meter		1	7000

humidity sensor		1	7000
screening machine	500 kg/hr	1	25000
weight machine	0-100 kg	1	7000
stiching machine		1	7000
Camera		1	10000
Tank	2 ton each	4	Available at site
Godown	500 sq feet	1	Available at site
Gate		1	Available at site
Boundry	1500 sq feet	1	Available at site
Open shade	500 sq feet	1	Available at site
Flooring		1	Available at site



Layout of mini unit

10. We have customized our innovations:

- **Super Absorbent Polymer:** Developed using lignin and acrylic acid, this polymer exhibits an impressive water- holding capacity of 268 times its weight. It can retain water for a minimum of 35days from the day of irrigation.
- **Nobel Enzymes:** These enzymes are pH-independent and aid in the degradation of biomass. They contribute to the enhancement of microorganism populations, humus content, organic matter, and carbon levels in the soil.
- **Green-Synthesized Nanoparticles:** These nanoparticles are synthesized using eco-friendly methods and confer thermophilic properties to microorganisms, allowing them to thrive at temperatures of upto 78 degrees Celsius. They also promote biomass degradation in the soil.
- **Anaerobic Bioreactor:** Our anaerobic bioreactor creates artificial climatic conditions for microorganisms and maintains suitable temperatures even during exothermic reactions in the fertilizer